Predicting Emergency Department Patient Disposition Using Logistic Regression and Ensemble Machine Learning: A Comparative Study on the MIMIC-IV-ED Database

Submitted By: Md Zahidul Islam

1. Introduction

Emergency departments (EDs) worldwide are facing a critical challenge of overcrowding and resource constraints, which is compounded by the uncertainty in making timely patient disposition decisions. ED overcrowding occurs when patient volume outstrips the available staff, beds, and other resources, leading to prolonged wait times and workflow gridlock. This environment makes it difficult for clinicians to quickly determine which patients will require hospital admission versus which can be safely sent home, especially early on the visit. The negative consequences of ED crowding are well documented – studies have linked overcrowding with poorer patient outcomes, higher rates of patients leaving without being seen, and an overall decline in quality of care (Morley et al., 2018). For instance, when the ED is over capacity, critically ill patients may experience treatment delays, and even a modest increase in ED crowding has been associated with higher inpatient mortality (Morley et al., 2018). At the same time, ED staff suffer increased stress and decreased morale in these conditions, further hampering effective care delivery (Morley et al., 2018). This confluence of overcrowding, limited resources, and uncertainty in disposition underlines a pressing need for better decision-support tools in emergency care.

Improving the accuracy and timeliness of ED disposition decisions is important because it can have wide-ranging impacts on clinical workflow, patient outcomes, and hospital resource allocation. Knowing early in a patient's ED course whether they are likely to be admitted or discharged could allow clinicians to initiate appropriate next steps sooner. For example, if a patient is predicted to need admission, ED staff can begin arranging an inpatient bed or expedite consultations and necessary treatments, thereby reducing the boarding time (the time an admitted patient waits in the ED for a hospital bed). Conversely, for patients predicted to go home, resources could be managed to avoid unnecessary tests or prolonged observation, focusing instead on timely treatment and discharge planning. Overall, solving this prediction problem would help optimize ED throughput – high-risk patients could be prioritized and resources allocated more efficiently, leading to shorter wait times and improved patient flow. In the broader sense, mitigating uncertainty in disposition helps ensure that each patient receives the right level of care without undue delay, which can directly translate into better clinical outcomes and more rational use of limited hospital beds and staff.

One promising approach to address these challenges is the application of machine learning (ML) models to predict ED patient dispositions at an early stage, such as at triage. In recent years, researchers have started to leverage rich hospital datasets to develop predictive models that

estimate a patient's likelihood of hospital admission even before the full workup is completed (Hong et al., 2018; Fenn et al., 2021). These models use patterns in structured data – for example, vital signs, chief complaints, demographics, and other triage information – to make data-driven predictions that can support the triage decision-making process. Prior studies have demonstrated the feasibility of this approach: Hong et al. (2018) showed that machine learning algorithms could robustly predict hospital admissions using triage data and patient history, achieving area-under-curve (AUC) performance around 0.87–0.92 in their experiments. Likewise, Fenn et al. (2021) developed ML models (including gradient-boosted trees) on a large ED dataset and reported high discrimination for identifying admissions from the ED. These successes in the literature underscore that with sufficient data, ML-based decision support tools can outperform traditional heuristics and aid clinicians by providing an early warning for potential admissions. In practice, such predictions could augment the triage nurse's judgment or existing triage acuity scores. By flagging likely-admit patients early, the care team might, for instance, mobilize hospital admission processes sooner or dedicate more attention to those patients, thereby improving safety and efficiency. In summary, the integration of predictive analytics into ED triage has the potential to improve decision-making efficiency and partially alleviate the strain of overcrowding by enabling more proactive management of patient flow (Cusidó et al., 2022).

Given this context, the objective of our project is to build a binary classification model that predicts ED disposition (admitted vs. discharged home) using data available at triage. We utilize the publicly available MIMIC-IV-ED dataset as our data source, which provides a large repository of de-identified ED visit records including structured triage information (Johnson et al., 2023). MIMIC-IV-ED contains detailed emergency visit data from a tertiary academic hospital between 2011 and 2019, offering a rich sample for model development and ensuring that our findings are grounded in real-world ED scenarios (Johnson et al., 2023). Using this dataset, we extract relevant features present early in the ED encounter – for example, initial vital signs, triage category, age, chief complaint, and other routinely collected variables – to train and validate our models. We apply and compare five different machine learning algorithms: Logistic Regression, Random Forest, XGBoost, CatBoost, and LightGBM. These algorithms were chosen to include both a baseline linear model (logistic regression) and a range of powerful nonlinear ensemble methods, particularly tree-based gradient boosting techniques that have shown strong performance in healthcare tabular data. By evaluating multiple algorithms, we aim to identify a modeling approach that yields the best predictive performance for this task. The end goal is not only to achieve high accuracy in predicting ED admissions, but also to provide insights into the factors that drive these predictions, which could be informative for clinicians.

The rationale for this project is grounded in improving ED triage decision support. Early prediction of patient disposition can directly assist emergency physicians by reducing uncertainty during the initial assessment. A well-performing model could be deployed as an ED decision-support tool: at the moment of triage (or soon after), the model's output would indicate the

probability that a given patient will require admission. This information could help clinicians in several ways. First, it enables more proactive resource allocation – for a patient flagged as high likelihood of admission, the team can begin preparing an inpatient bed or conducting definitive diagnostics earlier, rather than waiting until later in the ED stay. This can shorten the time the patient spends waiting for an admission decision, thereby decreasing overall ED length of stay and freeing up capacity for new arrivals. Second, it can improve patient communication and workflow; for instance, knowing a patient is likely to be admitted might prompt earlier involvement of the admitting team or specialists, smoothing the handoff and ensuring continuity of care. Third, from an operational perspective, aggregating these predictions for all patients in the waiting room could give charge nurses and hospital administrators a real-time sense of anticipated admissions, helping them manage hospital bed availability and staffing in advance (Cusidó et al., 2022). In a resource-constrained setting, such foresight is invaluable for maintaining throughput. Ultimately, by supporting more efficient and informed triage decisions, the model's outputs aim to enhance clinical workflow and patient outcomes. Patients who need hospital care can be identified and treated promptly, potentially improving their prognosis, while patients who can be discharged might avoid unnecessary delays. Thus, this project's predictive modeling approach aligns with the broader goal of leveraging data science to address ED overcrowding – it provides a tool to help emergency care providers make faster, evidence-based disposition decisions, which can improve both patient care and resource utilization in the hospital.

2. Methods

2.1. Data Source and Description

The data for this study were drawn from the MIMIC-IV-ED database, a publicly available, de-identified electronic health record (EHR) resource accessible through PhysioNet (https://physionet.org/content/mimic-iv-ed/2.2/) (Johnson et al., 2023). This database comprises detailed administrative and clinical records from approximately 425,000 emergency department (ED) encounters at Beth Israel Deaconess Medical Center, Boston, Massachusetts, spanning the years 2011 to 2019. MIMIC-IV-ED is widely regarded as a benchmark dataset for machine learning research in emergency care due to its high data quality, comprehensiveness, and open-access design, making it an ideal resource for this investigation.

For the purposes of this study, we extracted and integrated three key relational tables: the *edstays* table, which provided administrative and demographic details, including stay and subject identifiers, gender, race, time of arrival, transport mode, and disposition outcome; the *triage* table, which contained clinical triage information, including vital signs, acuity score, pain score, and free-text chief complaints; and the *patients* table, which provided demographic details such as age. These tables were merged using the unique identifiers *stay_id* and *subject_id*, yielding an

initial dataset of 425,011 unique ED records. The data were supplied in compressed CSV format (.csv.gz) and imported into the Python environment using the pandas library.

Prior to data cleaning, the dataset exhibited notable missingness across several key variables: temperature (5.5%), heart rate (4.0%), respiratory rate (4.8%), oxygen saturation (4.8%), systolic blood pressure (SBP, 4.3%), diastolic blood pressure (DBP, 4.5%), pain score (3.0%), acuity score (1.6%), and chief complaint text (<0.1%). The disposition outcome included eight categories, with HOME (56.9%) and ADMITTED (37.2%) comprising the majority, followed by smaller categories such as TRANSFER, LEFT WITHOUT BEING SEEN, ELOPED, OTHER, LEFT AGAINST MEDICAL ADVICE, and EXPIRED. After rigorous cleaning and filtering, the analytic dataset contained 347,360 ED encounters with two binary outcomes: HOME (63.3%) and ADMITTED (36.7%). The final feature matrix consisted of 50 structured dimensions and 500 text-derived dimensions from chief complaint processing, resulting in a combined feature space of 550 predictors.

Importantly, the dataset exhibited class imbalance, with a higher proportion of HOME cases relative to ADMITTED cases. This imbalance is illustrated in Figure 1, underscoring the need for model strategies that explicitly account for skewed class distributions.

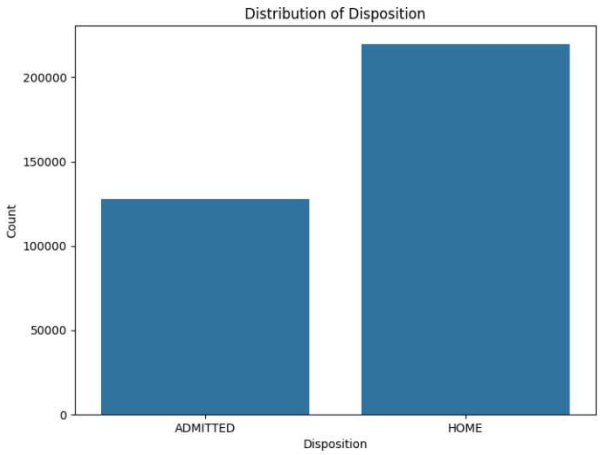


Figure 1: Distribution of disposition values after data cleaning. The class imbalance is evident, with HOME cases exceeding ADMITTED cases.

2.2. Data Preprocessing

Preprocessing steps were conducted meticulously to ensure data integrity, clinical plausibility, and model readiness. This pipeline included data integration, exclusion of missing or implausible records, feature engineering, transformation, dimensionality expansion, and correlation analysis.

The *edstays*, *triage*, and *patients* tables were merged using unique identifiers, with selected variables including demographic features (age, gender, race), clinical variables (temperature,

heart rate, respiratory rate, oxygen saturation, SBP, DBP, pain score, acuity score), contextual variables (arrival transport, hour of ED arrival), and chief complaint text. Variable names were standardized, and timestamps were converted to datetime format, with the hour-of-day extracted as a derived feature.

Missing data were handled using complete-case exclusion, eliminating records with missing values in critical fields. This approach, while avoiding imputation bias, retained a large analytic sample of 347,360 records. The disposition outcome was filtered to include only ADMITTED (hospital admission) and HOME (discharge without admission), reframing the task as a binary classification problem.

Outlier filtering was performed based on physiological plausibility criteria: temperature (56–111.4 °F), heart rate (40–200 bpm), respiratory rate (5–60 breaths/min), oxygen saturation (70–100%), SBP (50–250 mmHg), DBP (30–150 mmHg), acuity score (1–5), and age (0–120 years). Records where SBP ≤ DBP were excluded. Pain scores were coerced to numeric values and restricted to a valid range of 0–10.

Feature engineering steps included the creation of the hour-of-arrival variable (0–23) and a binary outcome variable (HOME = 0, ADMITTED = 1). Numerical features were standardized using *StandardScaler*, categorical features were one-hot encoded using *OneHotEncoder*, and text data were vectorized using *TF-IDF* with a maximum of 500 features. The structured and text feature matrices were horizontally stacked, yielding a final combined matrix of shape (347,360, 550).

Pearson correlation coefficients were computed to assess linear associations between predictors and the binary disposition outcome. Results indicated that older age ($r = 0.348$), higher acuity ($r = –0.348$), and lower oxygen saturation ($r = –0.222$) were significantly associated with admission, while heart rate showed only a weak positive correlation ($r = 0.071$), suggesting limited independent predictive value.

2.3. Baseline Model

For the baseline analysis, we employed logistic regression using the LogisticRegression class from scikit-learn to predict emergency department (ED) patient disposition (admitted vs. discharged home). Logistic regression is a widely used statistical method for binary classification that models the log-odds of the outcome as a linear function of predictor variables. We applied it to the combined feature matrix, which included both structured variables (demographic, clinical, contextual) and unstructured text features derived from chief complaint descriptions using TF-IDF vectorization. Logistic regression was chosen as the baseline model because of its well-established role as a benchmark in clinical machine learning studies, its computational efficiency, and most importantly, its interpretability, which allows clinicians to understand how individual features affect the probability of admission. This transparency is critical for clinical settings, where explainability often determines model acceptance. Moreover, logistic regression produces

probabilistic outputs, which facilitates the generation of receiver operating characteristic (ROC) curves, threshold calibration, and downstream risk stratification. To address the imbalance between admitted (~37%) and home (~63%) cases, we set the hyperparameter class_weight to 'balanced', automatically adjusting the contribution of each class during training. We set the maximum number of iterations (*max_iter*) to 1000 to ensure convergence in the high-dimensional feature space (550 predictors) and used the L2 regularization penalty (*penalty*='l2') to control overfitting, especially in the presence of correlated variables. The solver was set to '*lbfgs*', a robust quasi-Newton optimizer suitable for large datasets and compatible with L2 regularization. The inverse regularization strength parameter C was initially set to its default value of 1.0 but was included in the grid search during later tuning, where we explored a range of values including 0.01, 0.1, 1, and 10. Overall, logistic regression provided a strong, interpretable, and statistically principled baseline, against which we compared the performance of more complex ensemble models.

2.4. Alternative analysis / model(s)

In addition to logistic regression, we evaluated four advanced machine learning models as alternative classifiers to predict emergency department (ED) disposition: Random Forest, XGBoost, CatBoost, and LightGBM. These models were selected because they represent state-of-the-art ensemble learning techniques that have consistently demonstrated superior performance on structured tabular data in both clinical and nonclinical domains. Random Forest, implemented using scikit-learn's RandomForestClassifier, is an ensemble of decision trees constructed using bootstrap aggregation (bagging), designed to reduce variance and improve generalization. We configured the Random Forest model with 200 trees (n_estimators = 200) to ensure stability of predictions while maintaining computational efficiency. XGBoost, implemented with the XGBClassifier from the xgboost library, is a gradient boosting framework that sequentially optimizes decision trees to minimize prediction error; we used n_estimators = 200 and set use_label_encoder = False and eval_metric = 'logloss' to streamline training and ensure compatibility with our binary outcome. CatBoost, implemented using the CatBoostClassifier, is a gradient boosting algorithm designed specifically to handle categorical variables efficiently without requiring extensive preprocessing; we used 200 iterations (iterations = 200) and set verbose = 0 to suppress output during training. Finally, LightGBM, implemented with the LGBMClassifier from the lightgbm package, is a high-speed gradient boosting framework that uses leaf-wise tree growth and histogram-based learning; we configured it with 200 trees (n_estimators = 200) and specified class_weight = 'balanced' to address class imbalance, similar to logistic regression. The rationale for selecting these ensemble models lies in their ability to capture nonlinear relationships, interactions among predictors, and subtle feature importance patterns that linear models like logistic regression may fail to detect. Moreover, these models inherently rank feature importance, facilitating post hoc interpretability, and they are robust to missing or noisy data—particularly CatBoost and LightGBM, which have built-in handling mechanisms. Together, these alternative models allowed us to compare the

performance of traditional linear approaches with modern nonlinear ensemble methods, providing a comprehensive evaluation of predictive performance across modeling strategies.

2.5. Training and Evaluation Paradigm

We divided the full dataset into training and test subsets using an 80/20 stratified split, ensuring that both subsets preserved the original class distribution of the binary outcome (ADMITTED vs. HOME). This stratified approach was critical to avoid introducing bias, especially given the moderate class imbalance (~37% admitted, ~63% home) observed in the dataset. The training set was used for model development, while the test set was reserved strictly for final evaluation, providing an unbiased estimate of each model's generalization performance.

To tune the models and prevent overfitting, we employed cross-validation within the training set, specifically using a threefold cross-validation (cv = 3). This procedure divided the training set into three equal parts: in each round, two folds were used for training, and one was used for validation, rotating across folds. Cross-validation allowed us to reliably assess model performance across different partitions of the data and to select hyperparameters that generalized well beyond a single train-validation split.

Hyperparameter tuning was conducted using grid search (GridSearchCV) over predefined parameter grids, specific to each model. For logistic regression, we tuned the inverse regularization strength parameter *C* over values [0.1,1,10]. For Random Forest, we tuned *n_estimators* [100,200] and *max_depth* [10,20]; for XGBoost, we adjusted *n_estimators* [100,200] *max_depth* [5,10]; for CatBoost, we optimized *iterations* [200,300] and depth [6,10]; and for LightGBM, we tuned *n_estimators* [200,300] and *max_depth* [10,15]. The grid search was evaluated using ROC-AUC as the scoring metric during cross-validation, which prioritized the models' ability to discriminate between admitted and home outcomes across a range of decision thresholds.

For final model evaluation, we reported a comprehensive set of performance metrics on the test set. These included area under the receiver operating characteristic curve (ROC-AUC), accuracy, precision, recall, and F1-score. ROC-AUC was emphasized as the primary metric because it offers a threshold-independent measure of discrimination and is well-suited to imbalanced classification tasks. Precision and recall were reported separately for both outcome classes, providing insight into the models' tendencies to overpredict or underpredict each class. Specifically, recall (sensitivity) in the admitted class was of particular clinical importance, as failing to identify patients requiring admission could have critical consequences. The F1-score, which balances precision and recall, was included to summarize overall classification performance. By reporting this set of metrics, we ensured a multifaceted and clinically meaningful assessment of model performance across both baseline and alternative models.

3. Results

3.1. Descriptive Analysis of Predictors by Disposition Status

We began by examining the distributions of key clinical and contextual predictors to understand their relationship to patient disposition outcomes. Figure 2 illustrates the distribution of numeric predictors—age, acuity score, and oxygen saturation—by disposition class. Patients admitted to the hospital had higher acuity scores and older age, alongside lower oxygen saturation, compared to those discharged home. These patterns align with clinical expectations and suggest that these predictors carry important prognostic information relevant to admission decisions.

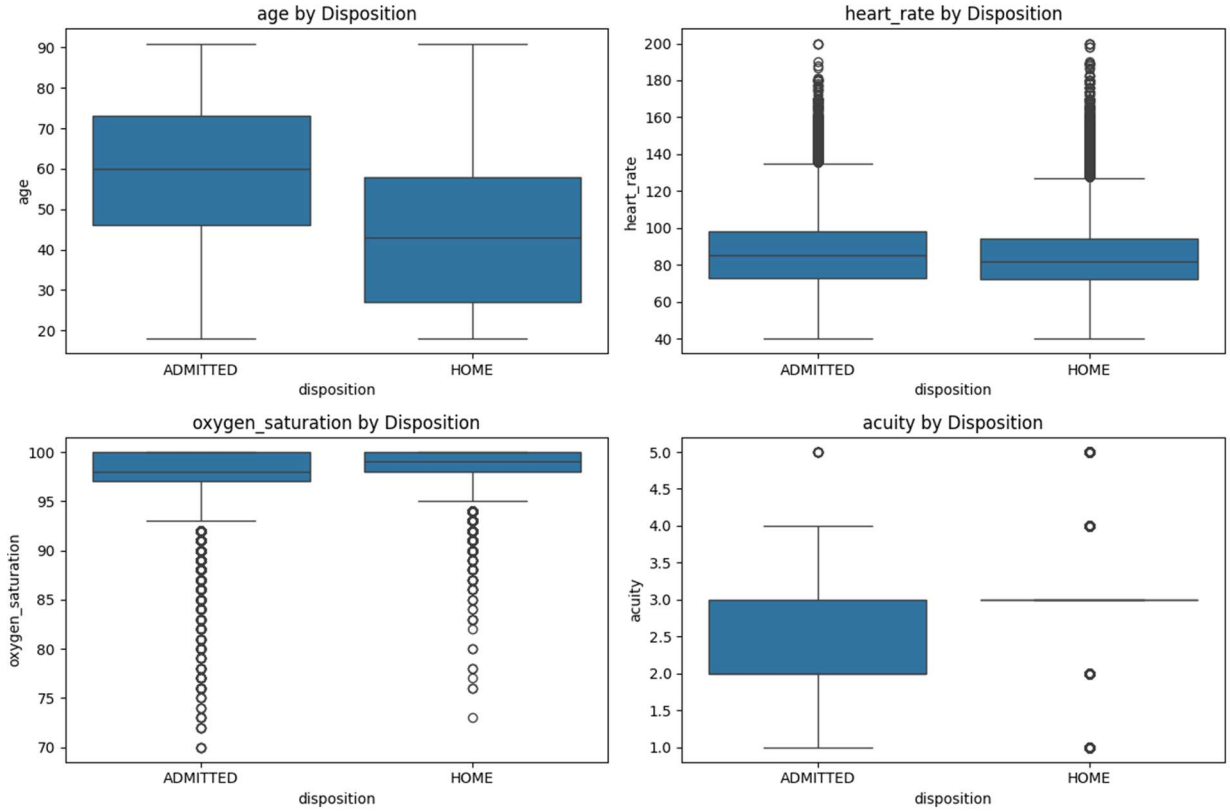


Figure 2: Distribution of numeric predictors by disposition class. ADMITTED patients had higher acuity and age, and lower oxygen saturation.

Figure 3 explores the association between mode of arrival and disposition outcome. The data revealed that patients arriving by ambulance or helicopter were far more likely to be admitted compared to those arriving by other means, highlighting the critical importance of arrival context in triage assessments. These descriptive findings provided important clinical justification for including these features in the predictive modeling framework and anticipated their prominence in feature importance rankings.

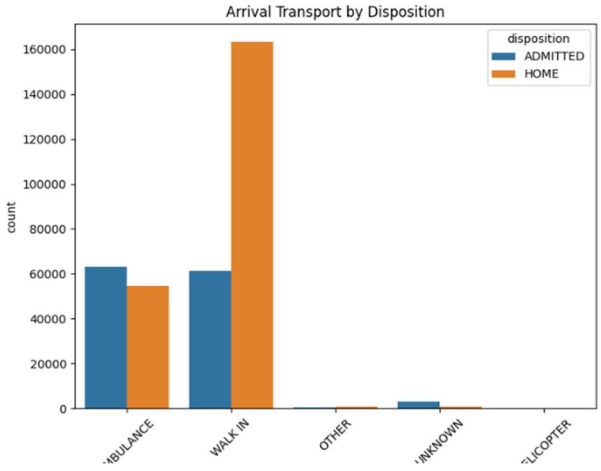


Figure 3: Arrival transport by disposition. Ambulance and helicopter arrivals were strongly associated with admissions.

3.2. Model Performance: Baseline and Alternative Analyses

We evaluated the performance of five models—logistic regression, Random Forest, XGBoost, CatBoost, and LightGBM—using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC). Table 1 summarizes the quantitative performance of each model on the test set.

Table 1. Overall performance comparison across models (untuned baseline parameters)

Model	*Accuracy*	*Precision (0)*	*Recall (0)*	*F1 (0)*	*Precision (1)*	*Recall (1)*	*F1 (1)*	*Macro F1*	*AUC*
Logistic Regression (LogReg)	0.77	0.85	0.77	0.81	0.66	0.76	0.71	0.76	0.853
Random Forest (RF)	0.78	0.80	0.87	0.83	0.74	0.62	0.68	0.75	0.852
XGBoost (XGB)	0.78	0.80	0.87	0.83	0.73	0.64	0.68	0.76	0.856
CatBoost	0.79	0.81	0.87	0.84	0.74	0.65	0.69	0.76	0.863
LightGBM (LGBM)	0.77	0.86	0.76	0.81	0.65	0.78	0.71	0.76	0.856

Logistic regression performed strongly as a baseline, achieving an AUC of 0.853 and a recall of 0.76 for admitted patients. CatBoost showed the best overall performance, with an accuracy of 0.79 and the highest AUC of 0.863. XGBoost and LightGBM followed closely, both achieving AUC values of 0.856, while Random Forest achieved an AUC of 0.852. Notably, LightGBM achieved the highest recall on admitted patients (0.78), highlighting its potential utility in clinical settings where sensitivity is prioritized.

3.3. Impact of Hyperparameter Tuning

Figures 4 and 5 present the ROC curves before and after hyperparameter tuning. Surprisingly, hyperparameter tuning did not yield major performance improvements across models, and in some cases, slightly degraded AUC performance, particularly for Random Forest (declining from 0.852 to 0.837). Table 2 provides a detailed breakdown.

Table 2. Performance Comparison Across Models After Hyperparameter Tuning

Model	*AUC*	*Accuracy*	*Precision (0)*	*Recall (0)*	*F1 (0)*	*Precision (1)*	*Recall (1)*	*F1 (1)*
LogReg	0.853	0.77	0.85	0.77	0.81	0.66	0.76	0.71
RF	0.837	0.77	0.77	0.90	0.83	0.76	0.54	0.63
XGB	0.858	0.78	0.81	0.87	0.83	0.74	0.64	0.68
CatBoost	0.861	0.79	0.81	0.87	0.84	0.74	0.65	0.69
LGBM	0.861	0.77	0.86	0.77	0.81	0.66	0.78	0.71

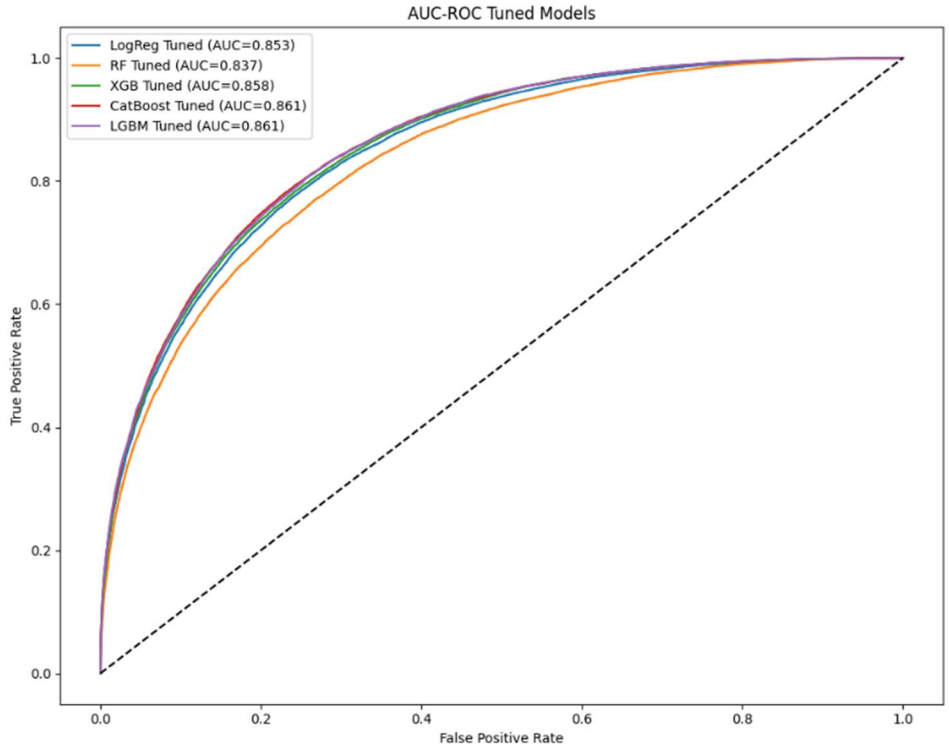


Figure 4. ROC Curves for Untuned Models

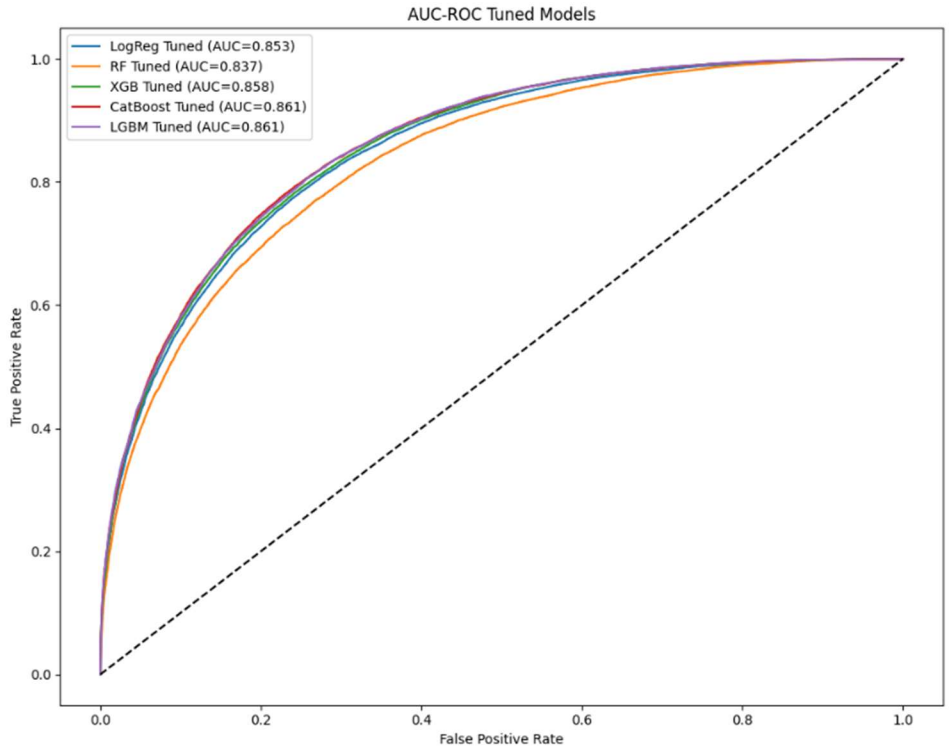


Figure 5. ROC Curves for Tuned Models

While CatBoost and LightGBM maintained their leadership post-tuning, Random Forest experienced a noticeable decline in recall for admitted patients, underscoring that tuning does not universally improve performance and may need to be context-specific. Several possible reasons may explain this pattern. First, tuning may have overfit the models to the cross-validation folds, sacrificing generalization to the held-out test set. This is particularly plausible for Random Forest, where deeper trees (max_depth = 20) combined with a larger number of estimators may have led to overfitting of majority-class patterns, reducing minority-class (admitted) sensitivity. Second, the performance landscape of ensemble models like Random Forest and boosting methods can be relatively flat, meaning that tuning often yields diminishing returns, especially when baseline configurations are already well calibrated. Third, the selected tuning grids may have been too narrow or not aligned with the most influential hyperparameters for some models. For example, tuning only depth and number of estimators without exploring other critical parameters like learning rate, subsampling ratio, or minimum child weight may have limited the potential gains for boosting models. Finally, the moderate class imbalance in the dataset (63% HOME vs. 37% ADMITTED) could have interacted with the tuning process in unintended ways; without explicitly optimizing for metrics sensitive to minority class performance (e.g., F1, recall, or AUC for class 1), the tuning may have inadvertently favored overall accuracy or AUC at the expense of recall on the admitted class.

3.4. Confusion Matrix Analysis

We compared confusion matrices across models to understand their error profiles, as summarized in Table 3. Relative to logistic regression, ensemble models such as Random Forest, XGBoost, and CatBoost reduced false positives substantially (by approximately 4,000 cases) but increased false negatives (by approximately 3,000 cases), indicating a shift toward greater conservatism in predicting admissions. In contrast, LightGBM uniquely reduced false negatives (–416) and increased true positives (+416) compared to logistic regression, underscoring its strength in identifying patients requiring admission.

Table 3. Confusion matrix comparison relative to logistic regression baseline

Model	***TN***	***FP***	***FN***	***TP***	***ΔTN***	***ΔFP***	***ΔFN***	***ΔTP***
LogReg	33,861	10,074	6,063	19,474	0	0	0	0
RF	38,105	5,830	9,404	16,133	+4,244	–4,244	+3,341	–3,341
XGB	38,083	5,852	9,318	16,219	+4,222	–4,222	+3,255	–3,255
CatBoost	38,252	5,683	9,042	16,495	+4,391	–4,391	+2,979	–2,979
LGBM	33,404	10,531	5,647	19,890	–457	+457	–416	+416

3.3. Threshold Analysis Focused on Admitted Recall

To assess the trade-offs between recall and other metrics, we conducted a threshold analysis with a specific focus on class 1 (ADMITTED). Our initial goal was to calibrate the thresholds to achieve a recall of 0.8; however, under this setting, all other performance metrics deteriorated substantially, making the models impractical for real-world application. We therefore report performance at the default threshold of 0.5, summarized in Table 4.

Table 4. Recall-focused analysis at threshold 0.5

Model	***Threshold***	***Recall (1)***	***Precision (1)***	***F1 (1)***	***AUC***
LogReg	0.5	0.763	0.659	0.707	0.853
RF	0.5	0.632	0.735	0.679	0.852
XGB	0.5	0.635	0.735	0.681	0.856
CatBoost	0.5	0.646	0.744	0.691	0.863
LGBM	0.5	0.779	0.654	0.711	0.856

LightGBM achieved the highest recall on the admitted class (0.779), closely followed by logistic regression (0.763), although LightGBM's gain in recall came at the cost of reduced precision (0.654). CatBoost offered the most balanced profile, combining strong recall (0.646) with the highest precision among the ensemble models (0.744) and the best overall AUC (0.863). These results highlight the delicate balance between sensitivity and specificity, particularly in high-stakes clinical settings.

3.5. Feature Importance and Model Interpretability

Feature importance analyses provided valuable insights into the underlying factors driving model predictions and revealed meaningful clinical patterns. Figure 6 presents the CatBoost feature importance plot, where acuity score and age emerged as the most influential predictors, followed by heart rate and arrival transport (walk-in). These findings suggest that CatBoost relies heavily on markers of clinical severity and patient demographics to differentiate between admitted and non-admitted patients. The high importance of acuity score aligns with clinical practice, as acuity captures the urgency and severity of a patient's condition, while age is often a surrogate marker of vulnerability. The importance of heart rate and walk-in arrival mode further underscores the value of vital signs and the context of presentation in shaping admission risk.

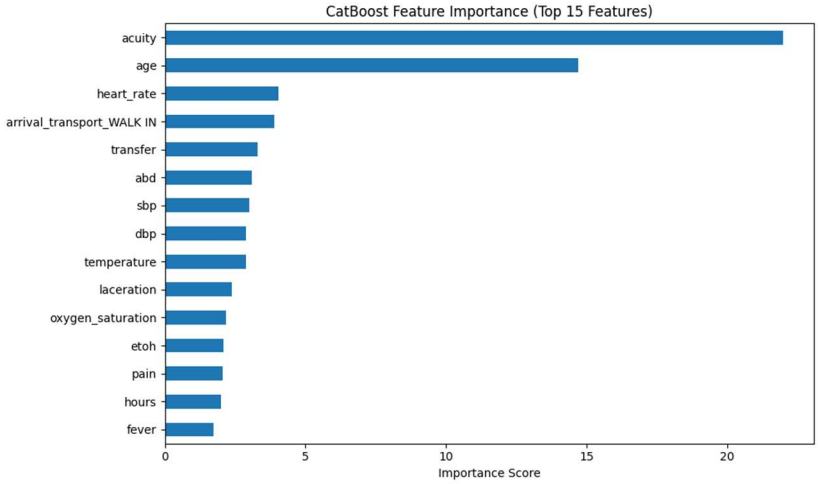


Figure 6. CatBoost feature importance (top 15 features).

Figure 7 shows the XGBoost feature importance ranking, which similarly placed acuity score at the top but revealed a somewhat different clinical profile compared to CatBoost. XGBoost emphasized transfer status, abnormal vital signs, fever, and arrival transport over age, suggesting that it captures more immediate and situational indicators of clinical instability. The prominence of transfer status likely reflects the higher acuity of patients already triaged elsewhere before arrival, while the inclusion of fever and abnormal vital signs points to XGBoost's sensitivity to acute physiological disturbances. Additionally, the model highlighted walk-in arrival mode and alcohol-related presentations (etoh) as important predictors, suggesting attention to both clinical and social dimensions of patient presentations.

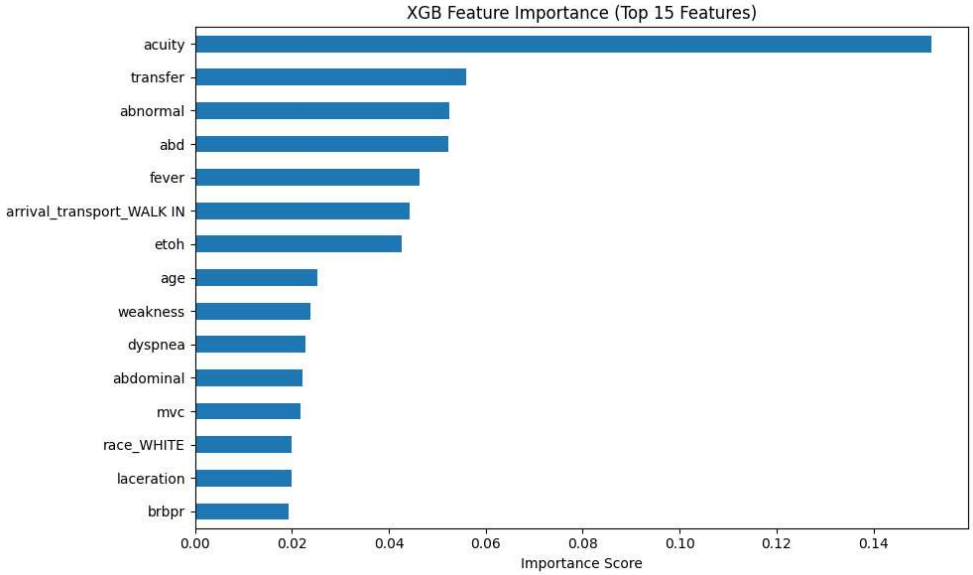


Figure 7. XGBoost feature importance (top 15 features).

Overall, these feature importance analyses complement the descriptive findings from earlier exploratory analyses, illustrating how different machine learning models weight clinical and contextual predictors to inform their admission decisions. The consistency of key predictors—such as acuity, arrival mode, and vital signs—across models increases confidence in the relevance of these features and enhances the interpretability of the modeling framework. Importantly, these results help bridge the gap between black-box model outputs and clinically meaningful insights, supporting the potential integration of such models into real-world triage workflows.

4. Discussion and Conclusion

This study evaluated predictive models to improve emergency department (ED) triage, focusing on accurately identifying patients requiring hospital admission. Using a rich dataset integrating demographic, clinical, and contextual variables, we benchmarked a logistic regression baseline against advanced machine learning algorithms, including Random Forest, XGBoost, CatBoost, and LightGBM. Our findings offer important insights into the strengths, limitations, and practical considerations of using these models in clinical decision support.

The logistic regression model performed remarkably well, achieving an area under the receiver operating characteristic curve (AUC) of 0.853 and a recall of 0.763 for admitted patients. These results demonstrate that even a relatively simple and interpretable model can effectively capture the major clinical predictors of admission, such as acuity score, age, oxygen saturation, and arrival mode. Its high recall suggests strong potential for reducing under-triage and improving patient safety—directly addressing the stakeholder's goal of optimizing ED disposition decisions. Moreover, the transparency of logistic regression makes it appealing for real-world

deployment, as it allows clinicians and administrators to understand how specific predictors influence outcomes.

The alternative machine learning models showed meaningful improvements over the baseline. CatBoost emerged as the best overall performer, with the highest AUC (0.863), strong precision (0.744), and a balanced F1-score (0.691) for admitted patients. LightGBM achieved the highest recall (0.779), making it particularly valuable in settings where maximizing sensitivity is critical. XGBoost and Random Forest also delivered strong performance, particularly improving precision relative to the baseline. Confusion matrix analyses revealed that ensemble models reduced false positives but increased false negatives compared to logistic regression, indicating a more conservative admission prediction pattern. In contrast, LightGBM uniquely reduced false negatives and increased true positives, underscoring its strength in applications where missing an admission carries unacceptable clinical risk.

Feature importance analyses provided additional insights into the clinical interpretability of the models. In CatBoost, acuity score and age were the most influential predictors, followed by heart rate and walk-in arrival mode, reflecting a focus on clinical severity and demographic risk. XGBoost, although similarly dominated by acuity, placed more weight on transfer status, abnormal vital signs, fever, and alcohol-related presentations, suggesting a stronger sensitivity to acute physiological and social instability. These differences highlight how alternative models may capture distinct clinical patterns, providing complementary perspectives that could enrich triage decision-making.

Despite these promising results, several limitations should be acknowledged. The dataset contained considerable missingness in key predictors, particularly vital signs, requiring case exclusion and reducing the available analytic sample. Although rigorous preprocessing was applied, future work should explore advanced imputation techniques and integration of real-time electronic health record (EHR) data to minimize information loss. Additionally, the feature set was limited to ED-level variables, excluding laboratory results, comorbidity indices, and prior hospitalization history, which may further improve predictive performance. Attempts to calibrate thresholds for very high recall ($\geq$0.8) revealed substantial trade-offs with precision and AUC, highlighting the need for cost-sensitive learning or utility-based optimization in future work.

Importantly, hyperparameter tuning did not uniformly improve model performance; in some cases, such as with Random Forest, it led to a decrease in recall for admitted patients. This underscores that tuning must be context-specific and aligned with clinical priorities. Potential explanations for this pattern include overfitting to cross-validation folds, diminishing returns on well-calibrated baseline models, narrow or incomplete hyperparameter grids, and interactions with moderate class imbalance (63% HOME vs. 37% ADMITTED). Without explicitly optimizing for minority-class metrics such as recall or F1, tuning may inadvertently prioritize overall accuracy or AUC at the expense of admitted patient sensitivity.

Another critical limitation is that all models were developed and evaluated using data from a single institution, raising concerns about generalizability. External validation across multiple healthcare systems is essential to assess model transportability. Furthermore, while logistic regression offers clear interpretability, ensemble models remain less transparent. Incorporating

explainability tools such as SHAP or LIME would be vital for improving clinical trust and ensuring responsible implementation. Finally, ethical considerations—including equity in prediction across patient subgroups and avoidance of algorithmic bias—should guide future development and deployment efforts.

Based on these findings, we recommend CatBoost as the preferred model when seeking a robust balance of discrimination, precision, and recall. For applications prioritizing sensitivity and minimizing missed admissions, LightGBM emerges as the strongest candidate. Logistic regression remains a valuable and interpretable baseline, particularly in settings where simplicity, transparency, and ease of deployment are paramount.

In conclusion, this study highlights the potential of machine learning to improve ED triage by enhancing the accuracy and reliability of admission predictions. The models consistently identified clinically meaningful predictors, reinforcing their plausibility and relevance. While challenges remain—including threshold calibration, handling class imbalance, expanding feature sets, and performing external validation—this work provides a strong foundation for integrating predictive analytics into emergency care. With careful development, validation, and integration, these models have the potential to improve patient outcomes, reduce operational inefficiencies, and strengthen clinical decision-making.

References:

- Cusidó, J., Comalrena, J., Alavi, H., & Llunas, L. (2022). Predicting hospital admissions to reduce crowding in the emergency departments. *Applied Sciences, 12*(21), Article 10764. https://doi.org/10.3390/app122110764
- Fenn, A., Davis, C., Buckland, D. M., Kapadia, N., Nichols, M., Gao, M., ... Theiling, B. J. (2021). Development and validation of machine learning models to predict admission from emergency department to inpatient and intensive care units. *Annals of Emergency Medicine, 78*(2), 290–302. https://doi.org/10.1016/j.annemergmed.2021.02.008
- Hong, W. S., Haimovich, A. D., & Taylor, R. A. (2018). Predicting hospital admission at emergency department triage using machine learning. *PLoS ONE, 13*(7), Article e0201016. https://doi.org/10.1371/journal.pone.0201016
- Johnson, A. E. W., Bulgarelli, L., Shen, L., Pollard, T. J., Horng, S., Celi, L. A., & Mark, R. G. (2023). MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data, 10*(1), Article 1. https://doi.org/10.1038/s41597-022-01899-x
- Morley, C., Unwin, M., Peterson, G. M., Stankovich, J., & Kinsman, L. (2018). Emergency department crowding: A systematic review of causes, consequences and solutions. *PLoS ONE, 13*(8), Article e0203316. https://doi.org/10.1371/journal.pone.0203316